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1 Introduction

The KrosFlo® Tangential Flow Filtration 2000 System (KTF-2000) provides a ready-to-use, flexible flow path solution to commercial-scale microfiltration and ultrafiltration process operations. The KTF-2000 system includes two magnetic drive centrifugal pump heads, dual custom flow path assemblies, flow meters, peristaltic pumps, KrosFlo® Hollow Fiber filters, and an optional connection for a weight scale for additional process monitoring.

Other components of the system include Retentate Flowmeters, Permeate Flowmeter, Pressure Sensors, Back Pressure Control Valves, Conductivity Sensor, Spectrum Control Panel controlled by an Allen Bradley PLC, an iFIX-based program to initiate actions and control operations, flexible tube flow paths, and a Spectrum stainless steel filter stand and lab cart that serve as the support structure.

2 Safety Precautions

2.1 Signal Words

The signal word and color distinguish the severity level of a hazard. Signal word definitions and colors are listed in TABLE 1.

TABLE 1. Signal Word Definitions and Colors

	Safety Alert Symbol (SAS) —used when a hazard to personnel is present. The SAS is omitted when the hazard is related to property/equipment damage only.
	A DANGER notification indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
	A WARNING notification indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
	A CAUTION notification indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.
	A CAUTION notification without the Safety Alert Symbol  indicates a potentially hazardous situation which, if not avoided, may result in property/equipment damage.

2.2 Safety Symbol Icons

The typical safety symbol icons—hazard, prohibition, and mandatory action—used on Spectrum systems are listed in TABLE 2. The icons are pictograms that communicate hazards quickly and across language barriers.

TABLE 2. Safety Symbol Icons

		
Electrical Hazard	Heavy Object	Heat Hazard
		
Radiation Hazard	Crush Hazard	Pinch Hazard
		
Inhalation Hazard	Authorized & Qualified Personnel Only	Read the Manual

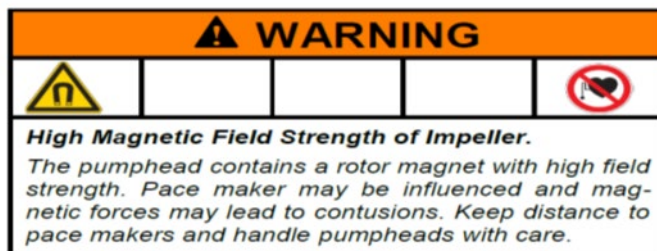
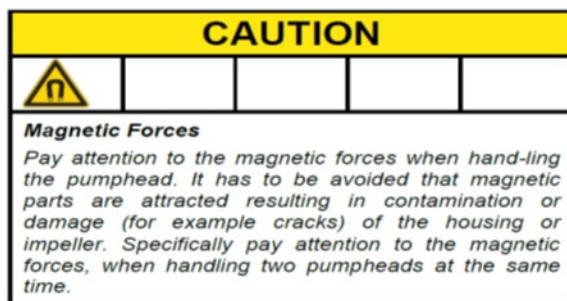
3 Safety Guidelines

To ensure safe operation:

1. Do not operate the system without first reading this manual.
2. Heed the following caution and warning statements:



WARNING – To avoid risk of electric shock, this equipment must only be connected to a supply mains with protective earth. No modification of this equipment is allowed.



WARNING – Do not plug in the system during Mechanical Assembly.



WARNING – Do not plug the system in during Electrical Connections procedure until instructed.



WARNING – System must be turned off when replacing fuses to prevent electrical hazard. Any electrical work beyond the electrical connections procedure and fuse replacement should be performed by trained personnel only.



WARNING – Peristaltic pumps must be turned off when replacing either tubing or pump heads to prevent crushing/pinching hazard.

4 System Specifications

TABLE 3. KTF-2000 System Specifications

Weight	300 lb. (approx.) 135 Kg		
Dimensions L x W x H	32 x 24 x 52.5 inches 81 x 61 x 133 cm	47 x 24 x 80 inches 120 x 61 x 203 cm (with extension)	
Power Requirements	200 - 240 VAC 1ph 15A (EU)	200 - 240 VAC 1ph 15A (USA)	
Outputs	Pumps: P-01 100 to 9000 RPM P-02 100 to 9000 RPM P-03 0.1 to 650 RPM P-04 0.1 to 650 RPM	Display: 15" LCD touchscreen	
Environmental	Electrical Cabinet and HMI – IP66	Operating Temperature: 0 to 50° C 32 to 122° F	
Chemical Resistance	Frame and Electrical Cabinet: SS-304	Castors: SS-304 and Polyurethane	Flow path Components: Polypropylene, polycarbonate, Polysulfone, and C-Flex/Pharmapure materials

5 Major Components

The major components used in the KTF-2000 TFF system are:

- **Retentate (Recirculating) Pump (P-01)** – Levitronix LPM-2000.7.10 low-shear centrifugal pump
- **Retentate (Recirculating) Pump (P-02)** – Levitronix LPM-2000.7.10 low-shear centrifugal pump
- **Buffer Pump (P-03)** – MasterFlex 77420-10 I/P peristaltic pump
- **Buffer Pump (P-04)** – MasterFlex 77420-10 I/P peristaltic pump
- **Retentate Flowmeters (FL-01, FL-02)** – Sonotec FS04.210 C0.55/230 V2.0
- **Permeate Flowmeter(FL-03)** – Levitronix LFC-1C-PC
- **Reactor Scale Input** – 4 to 20mA
- **Pressure Sensors** – Spectrum disposable pressure transducers
- **Controller** – Allen Bradley PLC (Programmable Logic Controller)
- **HMI** - Allen-Bradley Integrated Panel Display Computer (Human-Machine Interface)
- **Flow path** – Spectrum designed flexible tube flow path
- **Support Structure** – Spectrum stainless steel filter stand and lab cart

6 Instructions for Use

6.1 Installation and Mechanical Assembly

The KTF-2000 TFF system has a small footprint and requires only a small clean area (approx. 3m x 3m/10 ft. x 10 ft.) for assembly. No tools are needed for system assembly, though some basic hand tools such as screw gun, pliers, and screwdrivers are needed to uncrate the system. Some of the major components listed above are mounted on the cart. The other components can be removed from boxes and installed on the cart for use in the process.



WARNING – Do not plug in the system during Mechanical Assembly.

⚠ WARNING				
				
High Magnetic Field Strength of Impeller. <i>The pumphead contains a rotor magnet with high field strength. Pace maker may be influenced and magnetic forces may lead to contusions. Keep distance to pace makers and handle pumpheads with care.</i>				

CAUTION				
				
Magnetic Forces <i>Pay attention to the magnetic forces when hand-ling the pumphead. It has to be avoided that magnetic parts are attracted resulting in contamination or damage (for example cracks) of the housing or impeller. Specifically pay attention to the magnetic forces, when handling two pumpheads at the same time.</i>				

Follow these high-level steps to install and assemble the system:

1. Uncrate the system and remove all packaging material from the system components.
2. Locate and attach the proper filter brackets to the frame.
 The filter stand extension is attached to the cart frame by attaching it to the filter post. Be sure to place a black rubber gasket between the extension and frame post. Secure the extension with the provided metal clamp and tighten with a wrench.
3. Place the Buffer Pump (**P-03**) on the lower cart frame.
4. Place the Buffer Pump (**P-04**) on the cart frame.
5. The Recirculating Pumps (**P-01, P-02**) will each come already mounted to the filter cart.
6. Install and connect scale (**WE-01**) on the single-level cart.
7. Route and connect all cables. Cables are marked based on their destination. All cables should be routed to minimize kinking and obstructions.
8. Once the filter stand is assembled and pump motor is mounted, ready the pump head and flow path/MBT assembly items. Make sure all necessary components to complete the process line are available and ready (reactor, tubing, welding equipment, etc.). Please note that the process assembly should be designed for minimum hold-up volume.
9. To prepare the pump head and flow path/MBT assembly:
 - a. Carefully support the MBT/filter/pump head assembly and hold at an angle such that the pump head discharge lines up with the green color band at the pump support on the end of the motor drive.
 - b. Install the pump head into the pump motor drive making sure to disengage/pull out the locking pin to fully seat the pump head.
 - c. Rotate the pump discharge port and filter assembly to the vertical position and towards the red color band on the pump drive. The locking pin will click into place when the pump head is properly oriented in the pump drive.
 - d. Locate the filter in the 2 filter clamps on the frame post, making sure the clamps attach to the filter on the clear section of the housing (not on end fittings).
 Inspect the filter and pump before tightening clamps in place; there should be no strain placed upon any of the filter, pressure sensor or pump connections.
10. With filter and pump securely in place, install the permeate tubing, making sure there is no bending or kinking of the permeate tubing. Run extra tubing to the permeate vessel. Make sure permeate pressure sensors (**PE-03, PE-05**) are located between filter housing permeate port and any valves prior to permeate collection.

11. Locate the clamp-on retentate flowmeter (**FL-01, FL-02**) on each filter stand. Install the retentate tubing in the flowmeter and close the flowmeter clamp. Make sure there is 10 to 15 cm (4" to 6") of straight run tubing on each side of the meter. The return tubing may need auxiliary support to prevent kinking. Clamp the meter onto tubing. Note flow direction arrow on meter should point in up direction.

6.2 Electrical Installation

All electrical connections are made with plugs and sockets located on the rear side of the control panel.

Power Requirements for System

200 - 240VAC, 15 Amp Service, single phase.



WARNING – Do not plug in the system during Electrical Assembly until instructed.

To complete the system electrical connections:

1. Connect **P-01, P-02, P-03** and **P-04** connectors to panel.
2. Connect Flow Meter **FL-01, FL-02** and **FL-03** connectors to the flow meters.
3. Connect the pressure sensor cables (**PE-01 Feed, PE-02 Retentate, PE-03 Permeate, PE-04 Retentate**, and **PE-05 Permeate**) to the flow patch pressure sensors.
4. Connect **Main Panel power cord** to suitable electrical outlet (200 - 240VAC, 15 Amp Service).
5. Make sure that the **Emergency Stop (E-Stop)** button is pulled to **OUT** (inactive) position.
6. Turn on the Main Power by rotating the handle on the front of the control cabinet; this starts the computer and the control software loads automatically.
7. Press **Reset** button (Blue) on Control Panel.

7 KTF-2000 TFF System Operation

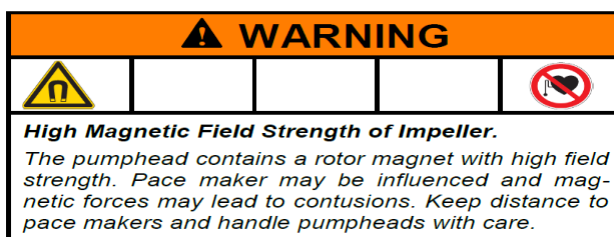
7.1 System Overview

The KTF-2000 operating system is an iFIX-based program installed on a touchscreen HMI PC. The program consists of six primary screens – Main (Operational), Settings, Chart (Process Trending), Alarm Setup, Alarm, and Report. Screen navigation is performed using the buttons on the bottom of each screen. Simply touch the button to go to the desired screen.

To end a session, navigate to the Settings Screen. Touch the **Exit iFIX** button located at lower right of screen, then close the iFIX program before turning off computer.

The HMI screen will go into screen saver mode while the system is operating or while program is open, just as a typical PC during times of inactivity. This will not affect operation and touching the screen will display the present active screen.

7.2 Getting Started



The KTF-2000 TFF System is ready to run after all Mechanical and Electrical Installation steps have been completed.

To run the KTF-2000 TFF System:

1. Pull the **E-Stop** button (located on front of Control Panel) to the **OUT** position.
2. Turn the front-panel **power switch** to **ON** position.
3. Press the **BLUE** reset button on control panel.

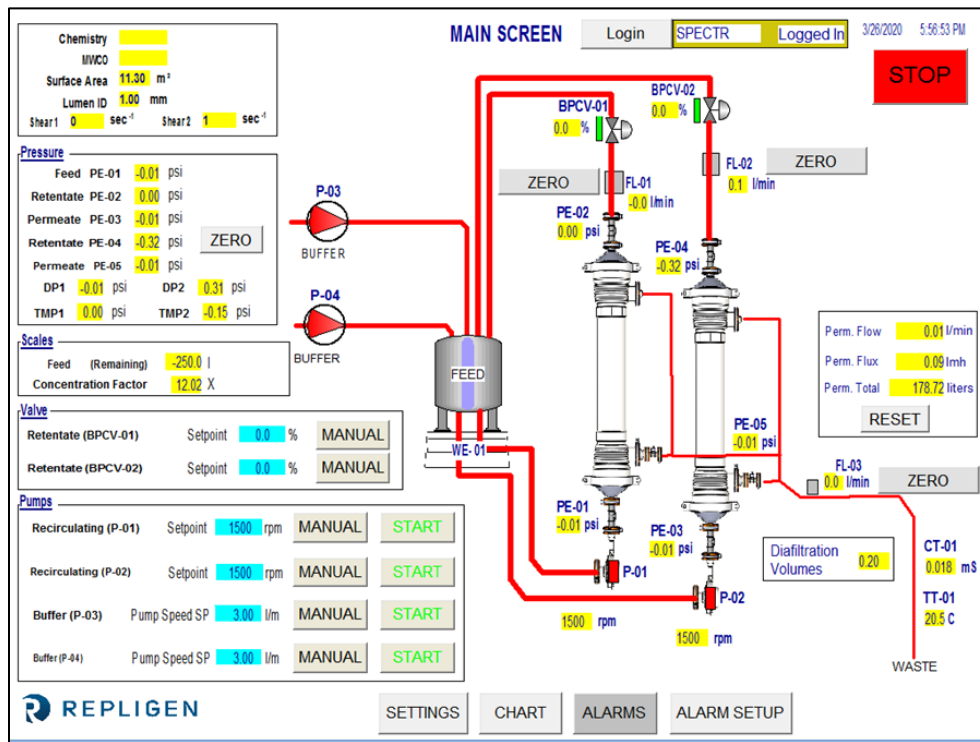
Once the computer is booted up, use the touchscreen to navigate to the **TFF** icon on the screen. Double click on the TFF icon to enter the KTF operating system. The control system will open to the Introductory screen, which contains the system part number and Installed Software Rev information. Touch this screen anywhere to navigate to the Main screen.

At startup, the KTF system will display all alarms as being active; they should be cleared before operating the KTF system. From the Main screen, navigate to the Alarms screen. Once at the Alarms screen, touch the **Reset Alarm** button on the touchscreen then touch the **Ack Alarms** button. All Alarms will be cleared.

NOTE: The **Blue** Reset button on the control panel enclosure is pressed to reset the Master Control Relay at startup, power outage, or E-Stop only. All other alarm conditions are reset from the Alarms screen on PC/HMI.

Be sure to exit iFix, and shutdown Windows before shutting down the KTF-2000 system.

7.3 Main Screen



Main Screen

The Main screen displays the KTF-2000 TFF System operational flow paths. The system operation and control features can be observed on the Main screen. Process data (flow, pressure, volume) is displayed on screen in real time. Process data OUTPUTS are displayed in the **YELLOW** highlighted boxes. Setpoint input data is displayed in the **BLUE** highlighted boxes.

Auto and Manual modes of operation for the Recirculating and Buffer pumps are selected here. The system can operate independently (outside of an automated process sequence), allowing the user greater flexibility in performing trials to optimize filter performance prior to entering a run.

Setpoint data is entered by touching a **BLUE** input box. Type in the new Setpoint data and press ENTER (= Equal Sign) button on the pop-up keyboard. The ENTER (= Equal) button MUST be pressed after data entry or the new Setpoint value will not be stored.

TABLE 4. Input Parameters for Main Screen

Parameter	Range	Units
BPVCs		
BPCV-01 Manual	0-100	%
BPCV-01 Auto	0-30	PSI
BPCV-02 Manual	0-100	%
BPCV-02 Auto	0-30	PSI
Pumps		
P-01 Manual Mode	0 – 9000	RPM
P-01 Auto Mode	0 – 75	LPM
P-02 Manual Mode	0 – 9000	RPM
P-02 Auto Mode	0 – 75	LPM
P-03 Manual (26 tubing)	0-4.06	LPM
P-03 Manual (73 tubing)	0-8	LPM
P-03 Manual (82 tubing)	0-13	LPM
P-03 Auto (26 tubing)	0-5000	Kg
P-03 Auto (73 tubing)	0-5000	Kg
P-03 Auto (82 tubing)	0-5000	Kg
P-04 Manual (26 tubing)	0-4.06	LPM
P-04 Manual (73 tubing)	0-8	LPM
P-04 Manual (82 tubing)	0-13	LPM
P-04 Auto (26 tubing)	0-5000	Kg
P-04 Auto (73 tubing)	0-5000	Kg
P-04 Auto (82 tubing)	0-5000	Kg

7.4 Automatic (Process) Mode

The KTF-2000 system is designed with an automated Process Sequence that controls a product concentration step followed by a diafiltration (wash) step and then a final concentration step.

The CONCENTRATION steps allow the system to run at a constant Transmembrane Pressure (TMP) and recirculation flow (P-01 or P-02 LPM) while concentrating the initial product sample volume to a predetermined volume reduction. The volume reduction, or Concentration Factor (CF) is determined by the user.

DIAFILTRATION sets the system to run at a constant Transmembrane Pressure (TMP) and recirculation flow rate (P-01 or P-02 LPM) while washing the concentrated sample product volume (Diafiltering) with buffer. The amount of wash (Diafiltration Volume, or DV) is determined by the user. **DV Setpoint** is used as the end point of the diafiltration sequence. This number signifies the number of diafiltration volumes that will be processed.

The FINAL CONCENTRATION step takes place once the diavolume setpoint has been reached for the diafiltration step. The concentration factor is set on the process Settings screen and can be observed from the Main screen.

The process steps are accessed on the Settings Screen (See section 7.6). A **Reset Windup** button is available on the Settings Screen to reset PID control parameters after the system has been stopped. This feature allows for a smoother restart of the system at auto process parameters. When the **Reset Windup** button is not pressed, the control loop will start at the pump speeds previously arrived at output.

7.5 Manual Mode

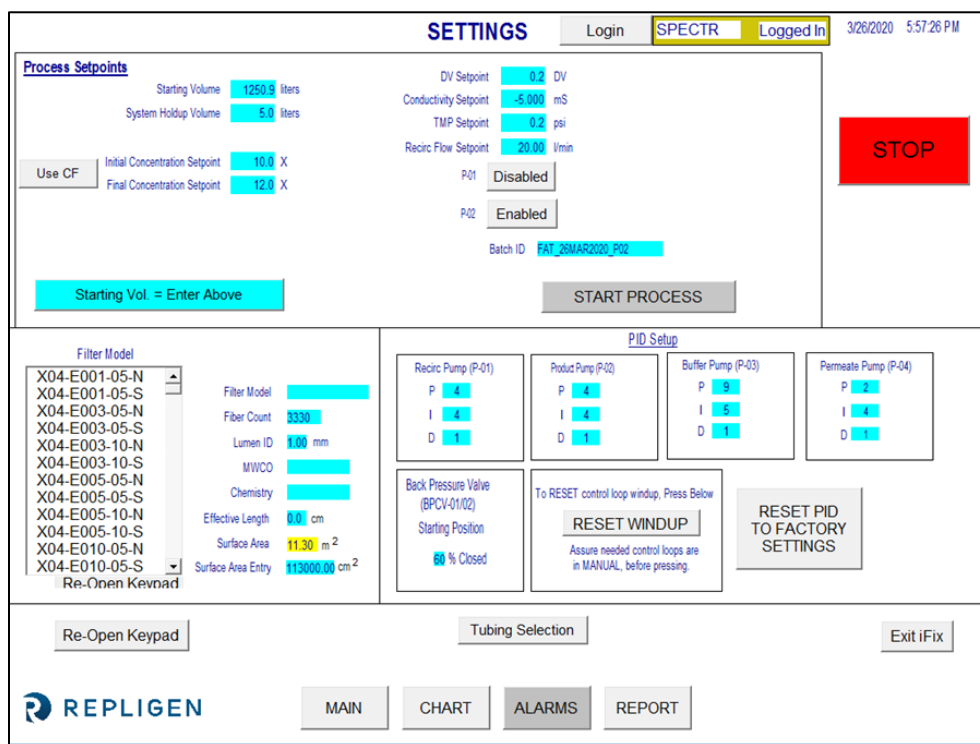
The KTF-2000 system can run in a manual mode which will operate the Back Pressure Control Valve (BPCV-01/02 % Closed), Buffer Pump (P-03/04 RPM) and Recirculation Pump (P-01/02 RPM) at constant settings with no process control.

The Manual mode button must be selected for each device (Pumps and Back Pressure Control Valves) and appropriate setpoints selected. Setpoints are entered in the **BLUE** boxes. Once started, the KTF-2000 system will maintain setpoint control, however in Manual mode, it will use a constant pump speed (RPM) or valve setting (% closed), and NOT a process variable control. Once started, the KTF-2000 system will maintain setpoint control during operation. Setpoints may be changed during TFF process operations.

7.6 Settings Screen

The Settings screen provides the **P-01 Enabled/Disabled** and **P-02 Enabled/Disabled** toggle buttons that control which flow path is to be used for TFF processes.

The Settings screen includes the settings for the **Filter Model**. Selecting a filter model will populate the **Fiber Count**, **Lumen ID**, and **Surface Area**. These parameters are used for the calculation of shear and flux. In the event a filter is not listed, the Fiber Count and Lumen ID must be entered in the **BLUE** boxes shown in the Filter Model box to ensure correct shear rate data.



The screenshot shows the 'SETTINGS' screen of the KTF-2000 system. At the top, there is a header with 'SETTINGS', 'Login', 'SPECTR', 'Logged In', and a timestamp '3/26/2020 5:57:26 PM'. Below the header, the screen is divided into several sections:

- Process Setpoints:** Includes fields for Starting Volume (1250.9 liters), System Holdup Volume (5.0 liters), Initial Concentration Setpoint (10.0 X), Final Concentration Setpoint (12.0 X), DV Setpoint (0.2 DV), Conductivity Setpoint (-5.000 mS), TMP Setpoint (0.2 psi), and Recirc Flow Setpoint (20.00 l/min). There are also toggle buttons for P01 (Disabled) and P02 (Enabled), and a Batch ID field (FAT_26MAR2020_P02).
- Filter Model:** A list of filter models (X04-E001-05-N to X04-E010-05-S) is shown on the left. On the right, fields for Filter Model, Fiber Count (3330), Lumen ID (1.00 mm), MWCO, Chemistry, Effective Length (0.0 cm), Surface Area (11.30 m²), and Surface Area Entry (113000.00 cm²) are displayed.
- PID Setup:** This section contains four sub-sections for different pumps: Recirc Pump (P-01), Product Pump (P-02), Buffer Pump (P-03), and Permeate Pump (P-04). Each sub-section has fields for P, I, and D values. Below these, there is a 'Back Pressure Valve (BPCV-01/02)' section with a 'Starting Position' field and a '% Closed' button. A 'RESET WINDUP' button is also present, with a note: 'To RESET control loop windup, Press Below'. A 'RESET PID TO FACTORY SETTINGS' button is also available.
- Buttons:** A large red 'STOP' button is located on the right side. A 'START PROCESS' button is at the bottom center. Other buttons include 'Re-Open Keypad', 'Tubing Selection', 'Exit iFix', 'MAIN', 'CHART', 'ALARMS', and 'REPORT'.

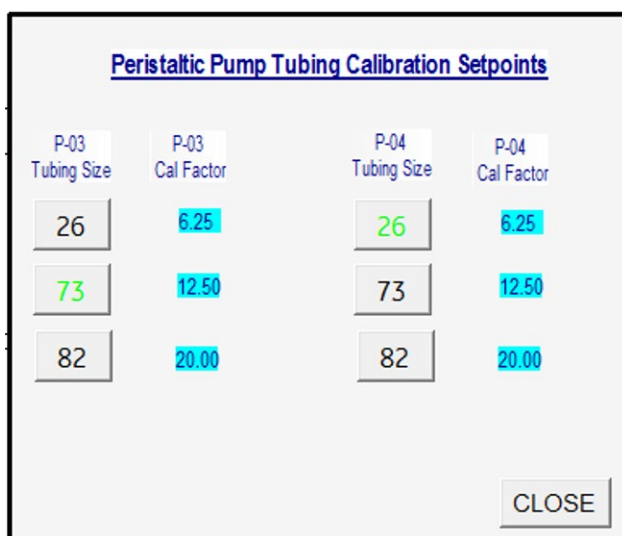
Settings Screen

The Settings screen also allows PID loop tuning of pump response for all pumps. The pump response to startup and setpoint change can be tailored using PID tuning. Setpoint Data is entered as described before: navigate to the appropriate Blue box and touch the setpoint box, type in data, and press ENTER (= equal sign) button on the pop-up keyboard.

All setpoint data should be entered prior to operation. Press **RESET PID TO FACTORY SETTINGS** to return all pumps to their default PID tuning values.

Press the **Use Weight/Use CF** toggle button to specify whether the Initial and Final Concentration setpoints are based on the specified weight (Kg) or concentration factor.

Press **Tubing Selection** to select the size of the tubing to be used with the KTF-2000 peristaltic pump (Product Pump (P-03/P-04)). You may also enter a calibration factor for the selected tubing on this popup screen.



Peristaltic Pump Tubing Calibration Setpoints			
P-03 Tubing Size	P-03 Cal Factor	P-04 Tubing Size	P-04 Cal Factor
26	6.25	26	6.25
73	12.50	73	12.50
82	20.00	82	20.00
CLOSE			

Tubing Selection Popup Screen

You may press the **Starting Vol = Enter Above/Starting Vol. = Use Scale** toggle button to specify whether the Starting Volume (that is, the volume in the feed tank at the start of the concentration process) is entered as a setpoint or is based on the weight of the tank (**WE-01**).

The process setpoints that may be configured from the Settings screen are defined as follows:

- **Starting Volume**—the volume in the feed tank at the start of the concentration process. This setting will be used to calculate the Concentration Factor, over time based on the remaining volume after the start of the process.
- **System Holdup Volume**—used to compensate for the volume of liquid that is in the flow path during the concentration process.
- **Initial Concentration Setpoint**—setpoint for the start of the Initial Concentration Sequence, if run.
- **Final Concentration Setpoint**—setpoint for the end of the Final Concentration Sequence.
- **DV Setpoint**—end point of the diafiltration sequence. This number signifies the number of diafiltration volumes that will be processed.
- **Conductivity Setpoint**—Value provides one of the end conditions for the Diafiltration sequence.
- **TMP Setpoint**—Constant Transmembrane Pressure setting for TFF processes.
- **Recirc Flow Setpoint**—controls the speed of the recirculation pump (P-01).

TABLE 5 lists the ranges of values for all input parameters on the Settings screen.

TABLE 5. Input Parameters for Settings Screen

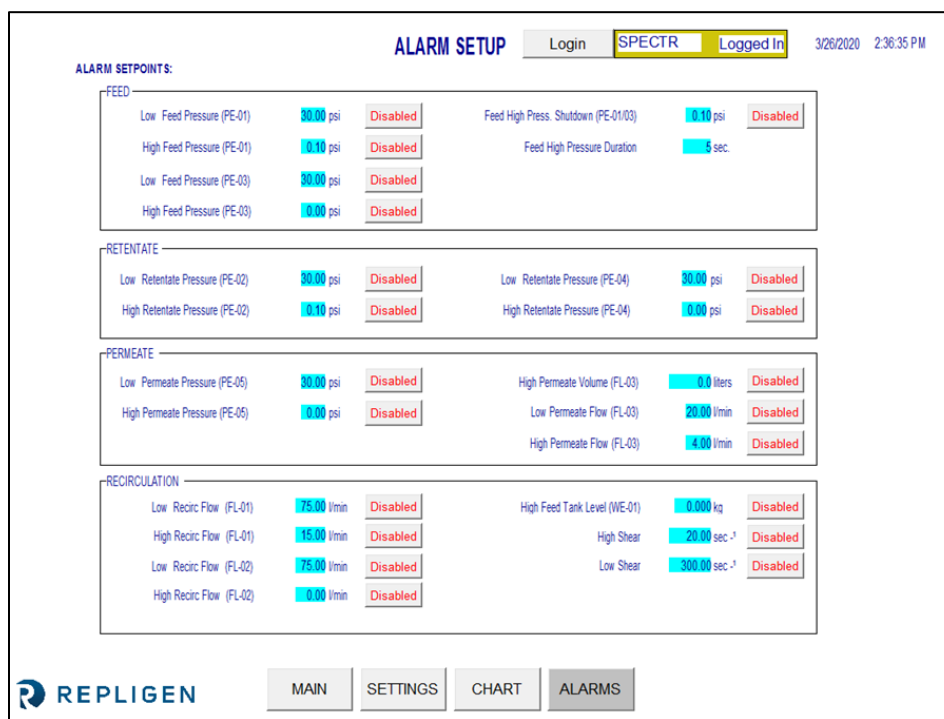
Parameter	Range	Units
Process Setpoints		
Starting Volume	0 - 1600	Liters
System Holdup Volume	0 - 100	Liters
Initial Concentration Setpoint	0 – 5000/0 - 100	Kg/X
Final Concentration Setpoint	0 – 5000/0 - 100	Kg/X
DV Setpoint	0 - 100	DV
Conductivity Setpoint	0 - 100	mS
TMP Setpoint	0 - 30	PSI
Recirc Flow Setpoint	0 - 100	LPM
Batch ID	Up to 20	Characters

Filter Model		
Filter Model Number	Up to 13	Characters
Fiber Count	0—50,000	
Lumen ID	0—100	mm
MWCO	Up to 10	Characters
Chemistry	Up to 10	Characters
Effective Length	Up to 5	Characters
Surface Area Entry	Up to 200,000	CM ²
Qty of Modules	Up to 1	Characters

PID Setup		
	Range	Default
Recirculation Pump (P-01)		
Recirculation Pump (P-01) P	0 to1000	4
Recirculation Pump (P-01) I	0 to1000	4
Recirculation Pump (P-01) D	0 to1000	1
Product Pump (P-02)		
Product Pump (P-02) P	0 to1000	4
Product Pump (P-02) I	0 to1000	4
Product Pump (P-02) D	0 to1000	1
Buffer Pump (P-03)		
Buffer Pump (P-03) P	0 to1000	9
Buffer Pump (P-03) I	0 to1000	5
Buffer Pump (P-03) D	0 to1000	1
Permeate Pump (P-04)		
Permeate Pump (P-04) P	0 to1000	9
Permeate Pump (P-04) I	0 to1000	5
Permeate Pump (P-04) D	0 to1000	1

Press the **Exit iFix** button located on the lower right corner of the Settings screen to exit the program.

7.7 Alarm Setup Screen



ALARM SETUP Login **SPECTR** Logged In 3/26/2020 2:36:35 PM

ALARM SETPOINTS:

FEED

Low Feed Pressure (PE-01)	30.00 psi	Disabled	Feed High Press. Shutdown (PE-01/03)	0.10 psi	Disabled
High Feed Pressure (PE-01)	0.10 psi	Disabled	Feed High Pressure Duration	5 sec.	
Low Feed Pressure (PE-03)	30.00 psi	Disabled			
High Feed Pressure (PE-03)	0.00 psi	Disabled			

RETENTATE

Low Retentate Pressure (PE-02)	30.00 psi	Disabled	Low Retentate Pressure (PE-04)	30.00 psi	Disabled
High Retentate Pressure (PE-02)	0.10 psi	Disabled	High Retentate Pressure (PE-04)	0.00 psi	Disabled

PERMEATE

Low Permeate Pressure (PE-05)	30.00 psi	Disabled	High Permeate Volume (FL-03)	0.0 liters	Disabled
High Permeate Pressure (PE-05)	0.00 psi	Disabled	Low Permeate Flow (FL-03)	20.00 l/min	Disabled
			High Permeate Flow (FL-03)	4.00 l/min	Disabled

RECIRCULATION

Low Recirc Flow (FL-01)	75.00 l/min	Disabled	High Feed Tank Level (WE-01)	0.000 kg	Disabled
High Recirc Flow (FL-01)	15.00 l/min	Disabled	High Shear	20.00 sec ⁻¹	Disabled
Low Recirc Flow (FL-02)	75.00 l/min	Disabled	Low Shear	300.00 sec ⁻¹	Disabled
High Recirc Flow (FL-02)	0.00 l/min	Disabled			

REPLIGEN MAIN SETTINGS CHART ALARMS

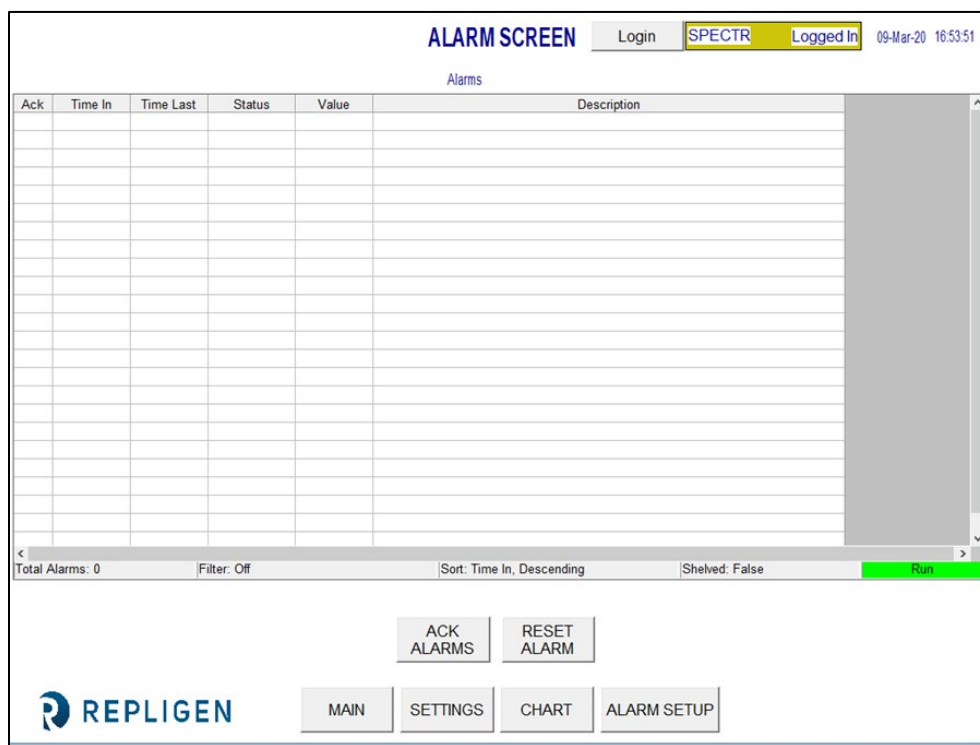
Alarm Setup Screen

The Alarm Setup screen contains 22 process alarm conditions to help protect the process from unfavorable process conditions. 21 of the Alarms have the option of being Enabled or Disabled. The one alarm condition which cannot be disabled is Feed High Pressure duration because it describes how long an overpressure condition can exist without setting the shutdown alarm.

TABLE 7. Alarm Setup Screen Setpoint Range

ALARM SETPOINTS	RANGE	UNITS
Low Feed Pressure (PE-01)	-5 to 30	PSI
High Feed Pressure (PE-01)	-5 to 30	PSI
Low Feed Pressure (PE-03)	-5 to 30	PSI
High Feed Pressure (PE-03)	-5 to 30	PSI
Feed High Pressure Shutdown (PE-01/PE-03)	-5 to 30	PSI
Feed High Pressure Duration	5 to 300	Sec
Low Retentate Pressure (PE-02)	-5 to 30	PSI
High Retentate Pressure (PE-02)	-5 to 30	PSI
Low Retentate Pressure (PE-04)	-5 to 30	PSI
High Retentate Pressure (PE-04)	-5 to 30	PSI
Low Permeate Pressure (PE-05)	-5 to 30	PSI
High Permeate Pressure (PE-05)	-5 to 30	PSI
High Permeate Volume (FL-03)	0 to 10000	L
Low Permeate Flow (FL-03)	0 to 20.0	LPM
High Permeate Flow (FL-03)	0 to 20.0	LPM
Low Recirc Flow (FL-01)	0 to 75	LPM
High Recirc Flow (FL-01)	0 to 75	LPM
Low Recirc Flow (FL-02)	0 to 75	LPM
High Recirc Flow (FL-02)	0 to 75	LPM
High Feed Tank Level (WE-01)	0 to 5000	Kg
High Shear	0 to 12000	SEC ⁻¹
Low Shear	0 to 12000	SEC ⁻¹

7.8 Alarms Screen



Alarms Screen

This Alarms screen is for viewing, acknowledging, clearing, and resetting all Alarms. A list of all Alarms, Alarm Range, and Tag (sensor or device responsible for Alarm Setpoint) is provided in TABLE 7.

NOTE: For the initial start-up (plugging in system to power), or in the event of power outage or Emergency Stop, press the Blue Reset button on the control panel enclosure to reset alarms. All other alarm conditions are reset from the Alarms screen.

The **Ack Alarm** button will acknowledge the alarm and clear it from the screen, if the alarm is not currently active.

The **RESET ALARM** button will silence the alarm and reset the flashing ALARM button once the system is operating in the normal range.

Alarms can be enabled or disabled manually using the **ALARM SETUP** screen. To disable or prevent an alarm, select the Disabled button. 'Disabled' should be displayed for alarm to be disabled. Likewise, 'Enabled' will be displayed if the alarm is enabled.

High alarms will trigger when process value rises to SP or above; Low alarms will trigger when process value falls to SP or below. The system is equipped with a short delay to avoid Low Alarms during start-up conditions. Alarms have two categories, Warning Alarms and Shutdown Alarms.

7.8.1 Warning Alarms

Flashing Button and Beeper will sound when Alarm setpoint is reached. The system will continue to run during Alarm conditions and indicate active alarm. Alarms can be acknowledged and cleared when Alarm Conditions are satisfied (i.e. – system operation restored within Alarm boundaries).

7.8.2 Shutdown Alarms

The system will shut down when a Shutdown setpoint is reached. The only shutdown alarm is Feed High Pressure Shutdown (PE-01).

7.9 Chart Screen



Chart Screen

Select the Chart button on the Main screen to display a chart for closer monitoring of all pressure and TMP values. The Chart screen is set up to display the parameters listed in TABLE 8.

TABLE 8. Chart Screen Parameters

Tag #	Parameter	Units
FL-01/FL-02	Feed Flow Rate	LPM
FL-02	Permeate Flow Rate	LPM
PE-01	Feed Pressure	PSI
PE-02	Retentate Pressure	PSI
PE-03	Feed Pressure	PSI
PE-04	Retentate Pressure	PSI
PE-05	Permeate Pressure	PSI
P-01	Recirc Pump Setpoint	RPM
TMP	Calculated TMP	PSI
Shear Rate	Shear Rate	Sec ⁻¹

Data can be viewed on the chart in real time or reviewed from a previous time period. Some features of the chart may be customized using the chart menu. The chart menu can be accessed by placing cursor on the chart and using left double-click.

8 Operation

8.1 Running TFF Processes

NOTE: Enter all TFF parameters on the Settings screen and select all Alarm Settings on the Alarm Setup screen prior to starting a sequence. The system must be fully assembled, tanks and components in place, and ready to operate. Be sure to enable the appropriate Recirculating Pump (P-01 or P-02) from the Settings screen for the desired flow path.

The KTF-2000 system is designed to run controlled TFF (Tangential Flow Filtration) processes with minimal operator input. Process parameters for Retentate flow rate and Permeate Flow are entered on the Sequences screen. The Feed Vessel Volume and Permeate Volume setpoints are also entered on Sequences Screen. The Filter Model is selected from the installed database on the Settings Screen.

After input of all Setup data, navigate to the Main Screen to initiate a Process run. The preferred start-up technique is to start the TFF process with Pump P-01 (or P-02), in Manual Mode at setpoints below parameters of the selected Auto Mode.

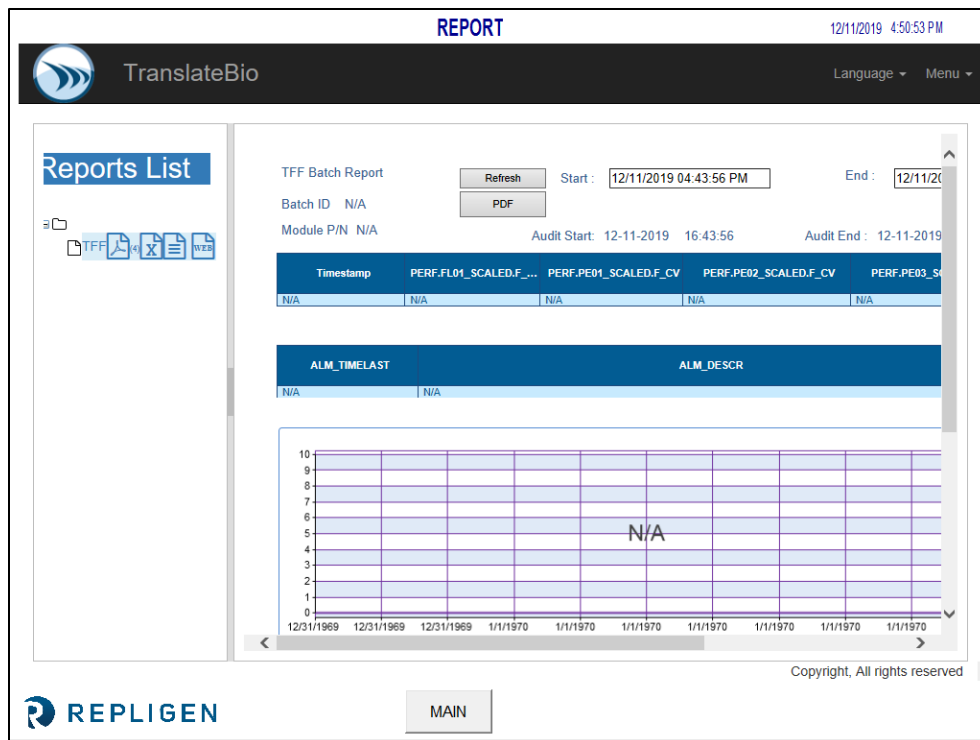
Once started, run the system in Manual Mode for a few minutes to completely flood the MBT and ensure air has been displaced. Once this is complete, stop the system and run the automated sequences from the Sequences Screen.

Once in Auto Mode, the KTF-2000 system will maintain setpoint control during operation. Setpoints can be changed while running during Auto operation.

Press **STOP** on the Settings or Main screens to stop the running process.

8.2 Generating Reports

Click **REPORT** on the Settings screen to access the **Report** screen.



Report Screen

The KTF-2000 continually logs collected data. Reports may be generated from this logged data as follows:

1. Click on the “WEB” icon under the ‘Reports List’. This will open the graphic shown above.
2. Enter the start time for the report in the **Start** field.
3. Enter the end time for the report in the **End** field.
4. Click **Refresh** to show the logged data collected over the specified time range displayed on the screen.
5. Click **PDF** to generate a portable document format (.PDF) of the report. This file will be opened when the report generation is done.
6. Click **MAIN** to return to the Main screen.

9 Safety

An Emergency Stop button (E-Stop) is located on the control panel enclosure. Pressing the E-Stop button will stop all operation immediately and prevent all components from operation. System operation can be restored by pulling the E-Stop out, pressing the Blue reset button, and then navigating to Alarms screen to Acknowledge and Reset the E-Stop Alarm. The desired sequence will have to be started/Restarted by the operator.

10 Maintenance

The KTF-2000 TFF system is of robust design and intended for use with process equipment. As such, lab or plant cleanliness and care are all that is needed to ensure quality operation when used in the system's intended manner. The frame, cabinet, and pumps can be cleaned by manual wipe-down using mild cleaning agents and/or warm water and a damp cloth or lab wipes. The HMI/Touchscreen should be cleaned with computer screen cleaner and computer screen wipes. Cleaning products are readily available from supply houses and the user must make sure the selected agent(s) are acceptable with their facility and compliant with system materials of construction as specified in Section 3. OEM manuals for major components are included in the documentation package for reference.

An Electrical Schematic and Installation List of all major components (Process and Control Panel items) is also provided should any service be required.



WARNING – System must be turned off when replacing fuses to prevent electrical hazard. Any electrical work beyond the Electrical Installation and fuse replacement should be performed by trained personnel.



WARNING – Peristaltic pumps must be turned off when replacing either tubing or pump heads to prevent crushing/pinching hazard.

11 Appendix

Contents:

11.1 Fuse Table

11.2 Installed Major Components

11.3 Recommended Spare Parts

11.4 Accessing the Scale Settings Screen

11.5 Normalized Water Permeability

11.1 Fuse Table

TABLE 9. System Fuses

Tag	Description	Fuse Type/Size
FU-1 (220 VAC Model)	Main Power 220/240 VAC	10A
FU-2	Empty	None
FU-2A	P-03	5A
FU-2B	Empty	None
FU-3	PS-2 Input Supply 120-240 VAC	5A
FU-4	PLC Input Supply 120-240 VAC	2A
FU-5	PS-1 Input Supply 120-240 VAC	5A
FU-6	PS-3 Input Supply 120-240 VAC	10A
FU-13	BPCV-01, BPCV-02	Buss AGC-2 2A
FU-17	PS-1 Output 24VDC	Buss AGC-5 5A

11.2 Installed Major Components

TABLE 10. Installed Major Components and Identification Tags

TAG	ITEM	MFG/MODEL
SYSTEM		
P-01	Recirculating Pump Motor	Levitronix LPM-2000.7.10
	Recirculating Pump Head	Levitronix DCP-2000.2
P-02	Recirculating Pump Motor	Levitronix LPM-2000.7.10
	Recirculating Pump Head	Levitronix DCP-2000.2
P-03	Product Pump Motor	MasterFlex 77420-10 I/P peristaltic pump
	Product Pump Head	MasterFlex
P-04	Product Pump Motor	MasterFlex 77420-10 I/P peristaltic pump
	Product Pump Head	MasterFlex
FL-01/FL-02	Retentate Flowmeter	Sonotec/Spectrum FS04.210 CO.55/230 V2.0
FL-03	Permeate Flowmeter/ Flow Converter	Levitronix LFC-1C-PC
PE-01/02/03/04/05	Pressure Sensors	Spectrum ACPM-799-01N
BPCV-01/02	Back Pressure Control Valve	Spectrum BPCV-82
CT-01	Conductivity Sensor	Hamilton Conducell 4USF
PANEL		
HMI	HMI/Computer	Allen Bradley 6181P15C2MWX1AC Integrated Panel Display Computer
PLC	PLC	Allen Bradley 1766-L32BWA MicroLogix 1400
PE-01 to PE-05	Signal Conditioner PE-01/PE-02/PE-03/ PE-04/PE-05	Allen Bradley 931S-B1C6D-DC
MCR-100	Master Control Relay	Allen Bradley 700-HF34Z24-4
	Ethernet Switch	N-Tron 104TX
PS-1	Power Supply 24 VDC	MeanWell SDR-240-24
PS-2	Power Supply 12 VDC	MeanWell MDR-20-12
PC-1/PC-2	Pump Controller	Levitronix LPC 2000.2-04

11.3 Recommended Spare Parts

The spare parts listed in Table 10 are recommended to prevent any downtime of the system should a minor failure occur.

TABLE 11. Recommended Spare Parts

Item	Description	Qty
1	FU-1, Main Power 220VAC	10A
2	FU-2, Spare	
3	FU-2A, Pump P-03	5A
4	FU-2B, Spare	
5	FU-3, PS-2 Input Supply 220VAC	5A
6	FU-4, PLC Input Supply 220VAC	2A
7	FU-5, PS-1 Input Supply 220VAC	5A
8	FU-6, PS-3 Input Supply 220VAC	10A
9	FU-13, BPCV-01/02	AGC-1 1A
10	FU-17, PS-1 Output 24VDC	AGC-5 5A
11	Spectrum Fittings Kit	1
12	1 ½" Tri-Clamp	2
13	1 ½" Tri-Clamp Gasket	5
14	1" Tri-Clamp Gasket	5
15	Permeate Tubing	5 m
16	Retentate Tubing	5 m

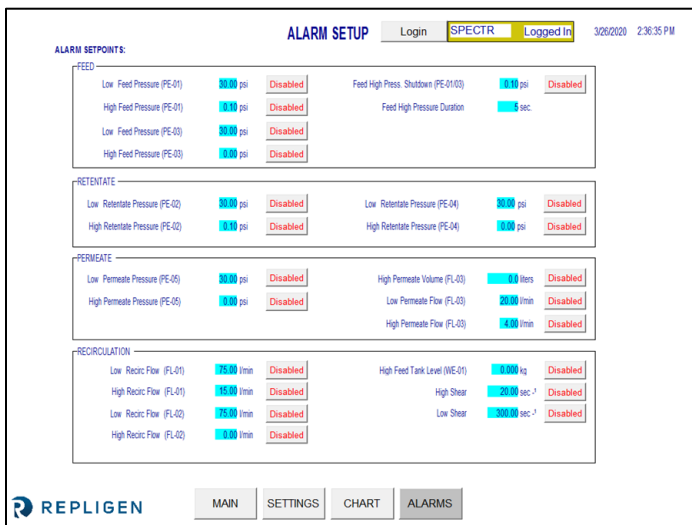
11.4 Accessing the Scale Settings Screen

The Scale Settings screen is a “hidden” screen the enables you to configure the maximum weight value for the scale (**WE-01**) connected to KTF-2000 system.

To access the Scale Settings screen and configure the maximum weight value for the scale (**WE-01**):

1. Touch the Repligen logo on the Alarm Setup screen.

Touch Repligen Logo



ALARM SETUP Login SPECTR Logged In 3/26/2020 2:36:35 PM

ALARM SETPOINTS:

FEED

Low Feed Pressure (PE-01)	30.00 psi	Disabled	Feed High Press. Shutdown (PE-01/02)	2.50 psi	Disabled
High Feed Pressure (PE-01)	10.00 psi	Disabled	Feed High Pressure Duration	1.00 sec	
Low Feed Pressure (PE-02)	30.00 psi	Disabled			
High Feed Pressure (PE-02)	10.00 psi	Disabled			

RETENTATE

Low Retentate Pressure (PE-02)	30.00 psi	Disabled	Low Retentate Pressure (PE-04)	30.00 psi	Disabled
High Retentate Pressure (PE-02)	10.00 psi	Disabled	High Retentate Pressure (PE-04)	10.00 psi	Disabled

PERMEATE

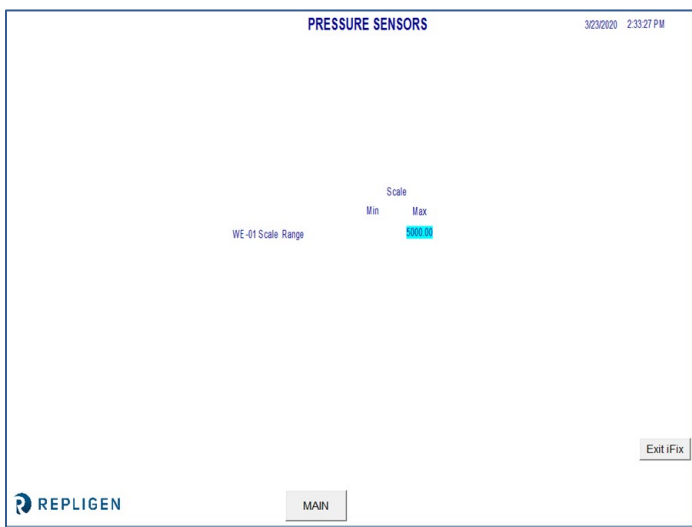
Low Permeate Pressure (PE-05)	30.00 psi	Disabled	High Permeate Volume (FL-03)	0.00 ltr	Disabled
High Permeate Pressure (PE-05)	10.00 psi	Disabled	Low Permeate Flow (FL-03)	20.00 l/min	Disabled
			High Permeate Flow (FL-03)	2.00 l/min	Disabled

RECIRCULATION

Low Racirc Flow (FL-01)	75.00 l/min	Disabled	High Feed Tank Level (WE-01)	0.00 kg	Disabled
High Racirc Flow (FL-01)	15.00 l/min	Disabled	High Shear	20.00 sec. ⁻¹	Disabled
Low Racirc Flow (FL-02)	75.00 l/min	Disabled	Low Shear	100.00 sec. ⁻¹	Disabled
High Racirc Flow (FL-02)	15.00 l/min	Disabled			

REPLIGEN MAIN SETTINGS CHART ALARMS

The Scale Settings screen appears.



PRESSURE SENSORS 3/23/2020 2:33:27 PM

Scale

Min Max

WE-01 Scale Range

0000.00

Exit iFix

REPLIGEN MAIN

2. Touch the **BLUE** field to configure the Maximum Value (0 to 600 Kg) for the **WE-01 Scale Range** setpoint.
3. Touch the **Main** button to return the Main Screen.